

Valuation of Fine Stringed Instruments

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INTRODUCTION

Fine stringed instruments — violins, violas, cellos, and bows — represent a specialised high-value asset class whose auction turnover reaches roughly \$100–300 million per year, supplemented by substantial dealer and private-sale activity. Owners (individuals, museums, art institutions) and insurers alike depend on accurate valuations, yet expert appraisals are costly and slow, and many instruments lack recent receipts. This project develops a **machine-learning regression model** that predicts fair market values from observable attributes — maker, instrument type, year and place of construction, and prevailing economic conditions — providing a fast, reproducible alternative to manual appraisal.

STAKEHOLDERS & KPI'S

Primary stakeholders are **private owners, museums, and art institutions** who need reliable valuations for sale or insurance purposes, and **insurance companies** seeking objective, data-driven price benchmarks. Two key performance indicators guide the project:

- Predictive accuracy on held-out sales, measured by R^2 and RMSE in log-price space.
- Practical applicability: the model must run in less than an hour and require only attributes available at the point of sale.

DATASET

Sales records were scraped from the **Tarisio / Cozio archive**, which aggregates results from 40 auction houses since 1829; 90 % of records post-date 1985. After filtering, the working dataset covers sales from **1965 to the present** (pre-1965 records were dropped due to unavailable CPI and FX data, affecting fewer than 5 % of entries). All prices were inflation-adjusted in local currency via CPI indices and then converted to real USD, making historical prices directly comparable. Faulty or incomplete entries were removed.

EXPLORATORY DATA ANALYSIS & FEATURE ENGINEERING

Because sale prices are heavily right-skewed, all modelling targets the **log-price**, which is approximately normal. The most important categorical feature is the **maker** (3 650 distinct values); the rarest 25 % of makers account for at most 24 sales each, motivating grouping by maker period and country/region. The type of instrument (taking 14 different values) is another important categorical feature. Beyond the raw catalogue attributes, the dataset was enriched with several self-constructed features:

- **Economic indicators** — volatilities, levels, returns and trend signals derived from S&P 500, gold, currency and yield data, plus a composite Market Climate Indicator (MCI) capturing the economic state around each sale date.
- **Geographic features** — country, region, and province of the instrument's city of origin, derived from the recorded maker city.

FEATURE & MODEL SELECTION

The cleaned dataset was split into **train, validation, and test sets** using a stratified split, ensuring a homogeneous presence of each type of instrument in the three splits.

Cardinality-aware encoding — low-cardinality categoricals (instrument type, currency) are one-hot encoded; high-cardinality ones (maker name, auction house, city, administrative region) are target-encoded to avoid dimensionality explosion.

A **LassoCV pipeline** (10-fold cross-validation) confirmed that **instrument type and maker** are the dominant price drivers, followed by economic indicators (S&P 500, gold) and geographic features. Features with zero Lasso coefficients were excluded from subsequent models. Five model families were then evaluated in a common preprocessing pipeline (median imputation, standard scaling, log-transform for skewed economic columns, one-hot / target encoding):

Model	RMSE (log)	R ² (log)	MAE (\$)	MAPE (log)
Local-Mean Baseline	0.710	0.712	12 689	6.2 %
Lasso Regression	0.688	0.729	14 132	6.1 %
Random Forest (200 trees)	0.639	0.767	12 029	5.6 %
LightGBM (tuned)	0.614	0.785	11 978	5.4 %
CatBoost (tuned)	0.611	0.787	12 252	5.4 %
Naive Ensemble (avg RF+LGBM+Cat)	0.610	0.787	11 922	5.4 %
Weighted Ensemble (Ridge stacking) ★	0.608	0.789	11 970	5.4 %

All metrics are evaluated on the held-out **test set**. LightGBM and CatBoost were tuned via 5-fold RandomizedSearchCV. The weighted ensemble stacks the three tree-based models with a Ridge meta-learner; optimised weights place ≈49 % on LightGBM and ≈51 % on CatBoost (Random Forest weight ≈0).

ERROR ANALYSIS

A structured error analysis of the best single model (CatBoost) across instrument types and countries of origin revealed that **prediction errors are largest for the rarest and highest-priced lots**: instruments by celebrated Italian makers sit at the tail of the price distribution and are systematically under-predicted due to sparse training examples. Errors are more moderate for common categories (modern factory instruments, bows) with denser archive coverage. **MAPE in log-space across the full test set is 5.4 %**, confirming strong average performance, while the high dollar-RMSE (~\$161 k) reflects the model's difficulty with six- and seven-figure outliers.

SUMMARY & OUTLOOK

The project demonstrates that **gradient-boosted tree ensembles**, fed with carefully engineered economic and geographic features, substantially outperform both the naive mean-by-type baseline and Lasso regression. The best model (weighted ensemble, R²log = 0.789) supports faster, cheaper, and more transparent appraisals for owners and insurers alike. Key limitations and next steps:

- **No instrument-level identity**: the Tarisio sales and instrument archives are poorly linked; condition, provenance, and individual instrument history are largely absent and would significantly improve predictions if available.
- **Multimodal signals**: high-resolution photos or audio recordings could be exploited by a convolutional or audio neural network to capture quality information invisible to tabular models.
- **Broader scope**: extending scraping to additional auction houses and integrating private-sale data would reduce distributional gaps for rare makers and instrument types.